Project Title:

Intelligent Test Case Generation and Defect Prediction Platform

Team Details

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Abstract:

Software testing is a critical phase of the Software Development Life Cycle (SDLC) that ensures software quality, reliability, and security. However, manually designing test cases and identifying defect-prone components require significant time and effort, often leading to incomplete test coverage and delayed software releases. **Intelligent Test Case Generation and Defect Prediction Platform** is an AI-driven solution that automates the software testing process by leveraging Machine Learning (ML), Natural Language Processing (NLP), and predictive analytics. The platform automatically generates optimized test cases from software requirements, user stories, and source code while simultaneously predicting defect-prone modules using historical bug data, code complexity metrics, and development patterns.

The proposed system helps developers and testers prioritize testing efforts by identifying high-risk areas early in the development cycle. It supports integration with modern DevOps and CI/CD workflows, enabling continuous testing and real-time quality assessment. By improving test coverage, reducing manual effort, and detecting defects before deployment, the platform enhances software quality, reduces development costs, and accelerates release cycles. This intelligent approach provides a scalable, efficient, and reliable solution for organizations seeking to improve software testing and quality assurance through automation and artificial intelligence.